

Transaction Management System

Overview

Develop a simple transaction management system consisting of an API for data operations and a frontend for displaying and adding transactions. The system will store transaction data in a CSV file.

Requirements

1. API Requirements

Create a backend API with two endpoints:

- **GET /transactions** – Retrieve all transactions from the provided CSV file
- **POST /transactions** – Add a new transaction to the CSV file

2. UI Requirements

Develop a web interface with:

- **Transaction Table** displaying:
 - Transaction Date
 - Account Number
 - Account Holder Name
 - Amount
 - Status
- **Add Transaction** button that opens a simple modal form with fields for:
 - Transaction Date (date input)
 - Account Number (text input)
 - Account Holder Name (text input)
 - Amount (number input)

3. Functionality Requirements

- Table should display all transactions retrieved from the API
- Adding a transaction via the modal should:
 - Submit data to the API endpoint
 - Randomly assign one of three statuses: **Pending**, **Settled**, or **Failed**
 - Update the table to show the new transaction

4. Data Format

- Transactions stored in a CSV file with columns:
 - Transaction Date (YYYY-MM-DD)
 - Account Number
 - Account Holder Name
 - Amount (decimal)
 - Status (Pending/Settled/Failed)

5. Technical Requirements

- **Backend:** Any framework
- **Frontend:** Any framework
- **Data Storage:** CSV file
- **Status Colors:**
 - Pending: Yellow
 - Settled: Green
 - Failed: Red

Implementation Notes

- No authentication required
- No complex form validation required, but it's nice to have.
- Libraries/components can be used freely
- Clear, step-by-step instructions on how to run the application. Write instructions suitable for someone unfamiliar with the framework or tools used. Do not make assumptions about prior knowledge.

Sample Data

Use this data as an initial sample

```
Transaction Date,Account Number,Account Holder Name,Amount,Status
2025-03-01,7289-3445-1121,Maria Johnson,150.00,Settled
2025-03-02,1122-3456-7890,John Smith,75.50,Pending
2025-03-03,3344-5566-7788,Robert Chen,220.25,Settled
2025-03-04,8899-0011-2233,Sarah Williams,310.75,Failed
2025-03-04,9988-7766-5544,David Garcia,45.99,Pending
2025-03-05,2233-4455-6677,Emily Taylor,500.00,Settled
2025-03-06,1357-2468-9012,Michael Brown,99.95,Settled
2025-03-07,5551-2345-6789,Jennifer Lee,175.25,Pending
2025-03-08,7890-1234-5678,Thomas Wilson,62.50,Failed
2025-03-08,1212-3434-5656,Jessica Martin,830.00,Settled
2025-03-09,9876-5432-1011,Christopher Davis,124.75,Pending
2025-03-10,4646-8282-1919,Amanda Robinson,300.50,Settled
```

Deliverables

1. Complete source code for both backend API and frontend application
2. Clear setup instructions
3. Running instructions with all necessary commands
4. Any configuration files needed

Evaluation Criteria

Your submission will be evaluated based on:

- Code quality and organization
- API design and implementation
- UI/UX design and functionality
- Error handling
- Documentation clarity
- Overall system architecture

Getting Started

When implementing this project, ensure your documentation includes:

- **Prerequisites**
 - List all required software and versions
- **Installation**
 - Step-by-step setup instructions
- **Configuration**
 - Any environment variables or config files needed
- **Running the Application**
 - Commands to start both backend and frontend
- **API Documentation**
 - Endpoint details with request/response examples
- **Testing**
 - How to test the application functionality



Please note:

- Write instructions suitable for someone unfamiliar with your chosen frameworks or tools. Do not make assumptions about prior knowledge.
- If you use AI tools while completing this task, please include a brief summary explaining how they were used and how they contributed to your solution. The use of AI is encouraged to help achieve the best possible result in the shortest amount of time, but it is not mandatory. Submissions completed with or without AI will be evaluated equally.

Good luck with your implementation!